

CSE 314 - Software Engineering Sessional

Online on Behavioral Patterns

Subsection: C2

Time: 25 minutes

You are building a financial dashboard for a day-trading firm. The traders need to see real-time changes in stock prices across various distinct widgets on their screen.

The core of the system is a StockData feed. Whenever the price of a specific stock changes, several unrelated components need to update immediately:

4. A **Ticker Tape** widget needs to scroll the new price.
5. A **Graph** widget needs to plot the new data point.
6. A **Buy/Sell Bot** needs to evaluate if the new price triggers an automated trade.

You need a design where the StockData feed doesn't know the specifics of these widgets but can notify all of them automatically whenever a price update occurs. Also there should be provision of adding or removing widgets

Task: Choose the appropriate design pattern to solve this problem and demonstrate with minimal code